RetainIQ

Explainable Customer Churn & Retention Intelligence Platform

1. Project Overview

RetainIQ is an end-to-end Machine Learning and Data Science project designed to help businesses identify customers who are likely to stop using their services and understand **why** they are at risk.

Instead of building only a basic churn prediction model, the project aims to develop a complete **Customer Retention Intelligence Platform** that combines:

- Data analysis
- Machine Learning
- Explainable AI
- Customer risk segmentation
- Retention recommendations
- Business analytics
- Interactive dashboards
- Backend APIs
- SQL database integration

The system will take customer demographic and behavioral data, predict the probability of churn, identify the major factors influencing the prediction, classify customers according to their risk level, and suggest appropriate retention actions.

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2. Problem Statement

Customer churn is a major business problem.

Companies may have thousands of customers but often cannot easily determine:

- Which customers are likely to leave?
- Which customers are at the highest risk?
- Why is a particular customer likely to churn?
- Which customers should the company prioritize?
- What retention action should be taken?

- What could be the potential business impact of retaining those customers?

A simple ML classifier that returns:

> "Churn" / "Not Churn"

does not provide enough information for practical decision-making.

RetainIQ aims to solve this by providing:

> **Prediction + Explanation + Prioritization + Recommended Action + Business Insight**

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3. Main Objective

Build an explainable ML-powered platform that can:

1. Analyze customer data.
2. Predict customer churn probability.
3. Identify high-risk customers.
4. Explain why a customer is considered high risk.
5. Segment customers according to churn risk.
6. Recommend appropriate retention actions.
7. Provide interactive business analytics.
8. Store customer and prediction information.
9. Allow individual and batch predictions.
10. Present everything through a professional dashboard.

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4. Machine Learning Objective

The core ML problem is:

Supervised Binary Classification

Given customer information:

Input:

- Customer demographics
- Tenure
- Satisfaction
- Order behavior
- Complaints

- Last-order activity
- Cashback
- App usage
- Other available customer attributes

Output:

```
Churn Probability
+
Risk Category
```

For example:

```
Customer ID: C10231

Churn Probability: 87%

Risk Level: CRITICAL
```

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5. Dataset

The initial implementation will use an e-commerce customer churn dataset containing customer demographic, behavioral, transactional, and service-related attributes.

Important types of features include:

- Tenure
- Satisfaction score
- Number of orders
- Days since last order
- Complaints
- Cashback amount
- App usage
- City tier
- Warehouse distance
- Coupon usage
- Customer demographics

The exact dataset structure and feature definitions will be analyzed before model development rather than blindly following a tutorial.

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6. Development Philosophy

This is the first major ML project being developed by the student.

Therefore, the project will **not be built by blindly copying an existing tutorial**.

The development approach will be:

```
Understand Concept
       ↓
Learn Required Theory
       ↓
Implement Concept
       ↓
Apply to Project
       ↓
Analyze Result
       ↓
Improve
       ↓
Continue
```

Whenever a new ML concept is required, it will first be understood conceptually and then implemented.

Examples include:

- Decision Trees
- Random Forest
- XGBoost
- ROC-AUC
- PR-AUC
- SMOTE
- Class weights
- Threshold selection
- ML Pipelines
- ColumnTransformer
- SHAP

The objective is to finish the project while also developing the ability to **explain every major technical decision during an interview**.

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7. Planned ML Development Pipeline

The project will progressively follow:

```
Problem Definition
        ↓
Dataset Understanding
        ↓
Data Cleaning
        ↓
Exploratory Data Analysis
        ↓
Feature & Target Identification
        ↓
Train/Test Split
        ↓
Baseline Model
        ↓
Logistic Regression
        ↓
Model Evaluation
        ↓
```

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8. Models

The project will progressively investigate several classification algorithms.

Baseline

Logistic Regression

Used as a simple and interpretable baseline.

Tree-based models

Decision Tree

Used to understand tree-based classification and model decision-making.

Random Forest

Used to understand ensemble learning and compare performance against a single tree.

Final candidate

XGBoost

Expected to be a strong candidate for the final model because of its effectiveness on structured/tabular data.

The final model will **not be selected simply because it has the highest accuracy**.

Model selection will consider appropriate classification metrics and the business objective.

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9. Model Evaluation

Because churn data can be imbalanced, accuracy alone will not be treated as the primary measure.

The project will evaluate models using metrics such as:

- Confusion Matrix
- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- PR-AUC

Particular attention will be given to **recall for churned customers**, since failing to identify a customer who is likely to leave can be more costly than incorrectly targeting a customer who would have stayed.

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10. Class Imbalance

Customer churn datasets commonly contain substantially more non-churn customers than churn customers.

The project will investigate approaches such as:

- Baseline model without balancing
- Class weights
- SMOTE
- Probability threshold adjustment

These approaches will be compared rather than blindly applying one technique.

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11. Explainable AI

A major differentiating feature of RetainIQ will be **Explainable AI using SHAP**.

Instead of only showing:

```
Churn Probability = 87%
```

the system should explain:

```
Why is this customer at risk?

↑ Low satisfaction
↑ Long time since last order
↑ Previous complaint
↑ Low order frequency

↓ Long customer tenure
```

This allows the user to understand the reasoning behind the model prediction.

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12. Customer Risk Segmentation

Customers will be categorized according to predicted churn probability.

Example:

```
0–25%       LOW
25–50%      MEDIUM
50–75%      HIGH
75–100%     CRITICAL
```

This allows businesses to prioritize their retention efforts instead of treating every customer equally.

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13. Retention Recommendation Engine

The platform will go beyond prediction.

Based on customer characteristics and identified risk factors, the system can recommend possible retention actions.

Example:

```
Customer Risk:
CRITICAL
```

```
Primary Risk Factors:
- Low satisfaction
- Long inactivity period
- Previous complaint

Suggested Actions:
- Resolve customer complaint
- Launch re-engagement campaign
- Provide personalized retention offer
```

The initial recommendation engine can use transparent business rules rather than an unnecessarily complicated AI system.

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14. Customer 360° View

Each customer can have an individual profile containing:

- Customer information
- Churn probability
- Risk level
- Behavioral information
- Relevant metrics
- Risk factors
- SHAP explanation
- Recommended retention actions

Example:

```
Customer C10231

Churn Probability: 87%
Risk: CRITICAL

Tenure: 3 months
Orders: 2
Satisfaction: 2/5
Complaints: Yes
Last Order: 21 days ago

Top Risk Factors:
1. Long inactivity
```

```
2. Low satisfaction
3. Complaint history

Recommended Action:
```

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15. Batch Prediction

The platform should support uploading a customer CSV file.

Example:

```
customers.csv
        ↓
Upload
        ↓
ML Pipeline
        ↓
5,000 customers analyzed
        ↓
Risk scores generated
```

The system can then identify:

```
947 High/Critical Risk Customers
```

and allow the results to be viewed or exported.

This is more realistic than manually entering customer information one customer at a time.

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16. Interactive Dashboard

The frontend will be designed as a professional **business intelligence / SaaS-style dashboard**, rather than a simple prediction form.

Main dashboard

Possible components:

- Total customers
- Overall churn rate
- High-risk customers
- Critical-risk customers
- Estimated revenue at risk

- Churn trend
- Risk distribution
- Top churn factors
- Highest-risk customer table

Customer page

- Search
- Filtering
- Sorting
- Customer profiles
- Risk scores
- SHAP explanations

Analytics page

Interactive visualizations such as:

- Churn by satisfaction
- Churn by tenure
- Churn by order frequency
- Churn by complaints
- Churn by customer segment
- Risk heatmaps

Model page

- Model comparison
- Precision
- Recall
- F1
- ROC-AUC
- PR-AUC
- Confusion matrix
- ROC curve
- Precision-Recall curve

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17. Retention Simulator

An optional advanced feature will estimate the potential business impact of targeting high-risk customers.

Example:

```
Customers targeted: 500

Estimated customers retained: 100

Campaign cost: ₹50,000

Estimated retained revenue: ₹4,00,000

Estimated ROI: 8×
```

This connects:

```
Machine Learning
       +
Business Analytics
       +
Decision Making
```

rather than focusing only on model accuracy.

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18. Backend

The application backend will be developed using:

Python + Flask

Responsibilities include:

- Receiving prediction requests
- Loading the trained ML pipeline
- Running predictions
- Returning prediction probabilities
- Returning risk levels
- Providing customer data
- Connecting frontend with database
- Supporting batch prediction

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19. Database

The project will use SQL-based persistent storage, with **PostgreSQL as the planned database**.

Potential tables:

```
customers
predictions
retention_actions
model_versions
```

Example prediction record:

```
Prediction ID
Customer ID
Churn Probability
Risk Level
Prediction Date
Model Version
```

The student already has SQL knowledge; PostgreSQL-specific concepts will be learned when the project reaches the database stage.

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20. Frontend

The planned frontend stack is:

- HTML
- CSS
- JavaScript
- Interactive charting/visualization library

The interface should feel like a real analytics product rather than a basic student form.

Design priorities:

- Professional layout
- Responsive design
- Interactive charts
- Search and filters
- Customer drill-down

- Clear risk visualization
- Loading/error/empty states
- Minimal but useful animations

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21. Deployment

Deployment will be considered after the core ML system and application are working.

Docker is **not mandatory** for the project.

It may be introduced later if it provides a clear benefit for reproducible deployment.

The priority is:

> Understanding and correctly building the ML system is more important than adding infrastructure technologies for resume keywords.

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22. Technology Stack

Data Science / ML

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- XGBoost
- SHAP

Backend

- Flask
- Python REST/API concepts

Database

- SQL
- PostgreSQL

Frontend

- HTML
- CSS
- JavaScript

- Data visualization library

Optional

- Docker
- Cloud deployment

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23. Development Versions

The project will be developed incrementally.

Version 0 — Project Understanding

```
Problem
Business Context
Dataset
Features
Target
Success Criteria
```

Version 1 — Basic ML

```
Data Cleaning
EDA
Preprocessing
Train/Test Split
Logistic Regression
Evaluation
```

Version 2 — Model Development

```
Decision Tree
Random Forest
XGBoost
Model Comparison
```

Version 3 — Advanced ML

```
Class Imbalance
SMOTE
Class Weights
Threshold Selection
Feature Engineering
```

```
Cross-Validation
Hyperparameter Tuning
Pipeline
ColumnTransformer
SHAP
```

Version 4 — Application

```
Flask
ML API
HTML/CSS/JS
Individual Prediction
Batch Prediction
```

Version 5 — Full Platform

```
PostgreSQL
Customer Management
Prediction History
Dashboard
Customer 360°
Risk Segmentation
Retention Recommendations
```

Version 6 — Final Polish

```
Deployment
Documentation
Architecture Diagram
Screenshots
Demo
Testing
Resume Presentation
```

Docker remains optional and will only be added if it is worth the time.

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24. What Makes RetainIQ Different From a Basic Churn Project?

A basic student project:

```
Dataset
   ↓
Train XGBoost
   ↓
92% Accuracy
   ↓
Flask Form
   ↓
Churn / No Churn
```

RetainIQ aims to become:

```
Customer Data
      ↓
Data Analysis
      ↓
ML Prediction
      ↓
Churn Probability
      ↓
Risk Segmentation
      ↓
Explainable AI (SHAP)
      ↓
Customer 360°
      ↓
Retention Recommendation
      ↓
Business Analytics
```

Therefore, the project demonstrates not only machine learning, but the **complete journey from raw data to a usable ML-powered decision-support application**.

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25. Final Goal

The final project should allow the student to confidently explain:

> **What business problem are we solving?**

> **How did we understand and prepare the data?**

> **Why is this a classification problem?**

> **How does each model work?**

> **Why did we choose the final model?**

> **How did we handle class imbalance?**

> **How did we evaluate the model?**

> **Why did we choose the prediction threshold?**

> **How does SHAP explain the prediction?**

> **How does the ML model communicate with the Flask backend?**

> **How is data stored in PostgreSQL?**

> **How does the dashboard turn predictions into business decisions?**

The project is considered successful when the student can **build, demonstrate, and technically defend the system in a placement interview**, rather than simply reproduce a tutorial.

Project End Goal in One Line

> **RetainIQ is an explainable, end-to-end customer retention intelligence platform that predicts churn, explains customer risk, prioritizes at-risk customers, and converts ML predictions into actionable business insights.**